



4425 Manchester
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Kalamazoo, MI 49001
Phone 269 381-9666
Fax 269 381-9698
www.karlabs.com

City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

Attn : Ms. Amy Vail

KAR Project No. : 114811

Date Reported : 11/07/11

Date Activated : 11/02/11

Date Due : 11/16/11

Date Validated : 11/07/11

Project

Description : Analysis of four aqueous samples for ultra-trace mercury.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 114811. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, reading 'David R. Alkema'.

David R. Alkema
Laboratory Manager

KAR Laboratories, Inc. maintains Full Certification status for Bacteriology, Inorganics, Regulated Organics and Synthetic Organics through USEPA, Michigan Department of Public Health and Indiana State Department of Health. This report may only be reproduced in full and not without the written consent of City of Cadillac, Water Resources.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **114811**

Date Reported : **11/07/11**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Effluent"**

Sampled By : *AV of City of Cadillac, Water Resources*

Date Received : **11/02/11**

Sample Date : **11/01/11**

Sample Type : **aqueous**

Sample Time : **1009**

KAR Sample No. : **114811-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	11/03/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	11/04/11	PML	

Sample ID : **"Effluent, Field Duplicate"**

Sampled By : *AV of City of Cadillac, Water Resources*

Date Received : **11/02/11**

Sample Date : **11/01/11**

Sample Type : **aqueous**

Sample Time : **1010**

KAR Sample No. : **114811-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	11/03/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	11/04/11	PML	

Sample ID : **"Field Blank"**

Sampled By :

Date Received : **11/02/11**

Sample Date : **11/01/11**

Sample Type : **aqueous**

Sample Time :

KAR Sample No. : **114811-03**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	11/03/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	11/04/11	PML	

Sample ID : **"Trip Blank"**

Sampled By :

Date Received : **11/02/11**

Sample Date :

Sample Type : **Trip Blk**

Sample Time :

KAR Sample No. : **114811-04**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	11/03/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	11/04/11	PML	

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Attn : Ms. Amy Vail

KAR Project No. : 113397
Date Reported : 08/22/11
Date Activated : 08/11/11
Date Due : 08/25/11
Date Validated : 08/22/11

Project

Description : Analysis of four aqueous samples for ultra-trace mercury.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 113397. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,


David R. Alkema
Laboratory Manager

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **113397**

Date Reported : **08/22/11**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Effluent"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **08/11/11**

Sample Date : **08/10/11**

Sample Type : **aqueous**

Sample Time : **1121**

KAR Sample No. : **113397-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	08/17/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	08/18/11	PML	

Sample ID : **"Effluent, Field Duplicate"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **08/11/11**

Sample Date : **08/10/11**

Sample Type : **aqueous**

Sample Time : **1122**

KAR Sample No. : **113397-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	08/17/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	08/18/11	PML	

Sample ID : **"Field Blank"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **08/11/11**

Sample Date : **08/10/11**

Sample Type : **aqueous**

Sample Time :

KAR Sample No. : **113397-03**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	08/17/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	08/18/11	PML	

Sample ID : **"Trip Blank"**

Sampled By :

Date Received : **08/11/11**

Sample Date :

Sample Type : **Trip Blk**

Sample Time :

KAR Sample No. : **113397-04**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	08/17/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	08/18/11	PML	



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Attn : Ms. Amy Vail

KAR Project No. : 111726
Date Reported : 05/09/11
Date Activated : 05/03/11
Date Due : 05/17/11
Date Validated : 05/06/11

Project

Description : Analysis of four aqueous samples for ultra-trace mercury.

Dear Client,

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If you wish to contact us about this work please mention KAR Project No. 111726. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,


David R. Alkema
Laboratory Manager

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **111726**

Date Reported : **05/09/11**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Effluent"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **05/03/11**

Sample Date : **05/02/11**

Sample Type : **aqueous**

Sample Time : **1255**

KAR Sample No. : **111726-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	05/04/11	PML	
Mercury by EPA 1631	0.6	ng/L	EPA 1631	05/05/11	PML	

Sample ID : **"Effluent, Field Duplicate"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **05/03/11**

Sample Date : **05/02/11**

Sample Type : **aqueous**

Sample Time : **1257**

KAR Sample No. : **111726-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	05/04/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	05/05/11	PML	

Sample ID : **"Field Blank"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **05/03/11**

Sample Date : **05/02/11**

Sample Type : **aqueous**

Sample Time :

KAR Sample No. : **111726-03**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	05/04/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	05/05/11	PML	

Sample ID : **"Trip Blank"**

Sampled By :

Date Received : **05/03/11**

Sample Date :

Sample Type : **Trip Blk**

Sample Time :

KAR Sample No. : **111726-04**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	05/04/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	05/05/11	PML	



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Attn : Ms. Amy Vail

KAR Project No. : 111711

Date Reported : 05/09/11

Date Activated : 05/03/11

Date Due : 05/17/11

Date Validated : 05/06/11

Project

Description : Analysis of one aqueous sample for Mercury Minimization.

Dear Client,

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If you wish to contact us about this work please mention KAR Project No. 111711. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink that reads 'David R. Alkema'.

David R. Alkema
Laboratory Manager

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **111711**

Date Reported : **05/09/11**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one aqueous sample for Mercury Minimization.*

Sample ID : **"Influent"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **05/03/11**

Sample Date : **05/02/11**

Sample Type : **aqueous**

Sample Time : **1250**

KAR Sample No. : **111711-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 245.2	05/05/11	DBL	
Mercury, total, low level	<0.0002	mg/L	EPA 245.2	05/05/11	DBL	



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Attn : Ms. Amy Vail

KAR Project No. : 110542

Date Reported : 02/11/11

Date Activated : 02/08/11

Date Due : 02/22/11

Date Validated : 02/10/11

Project

Description : Analysis of four aqueous samples for ultra-trace mercury.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 110542. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, reading 'David R. Alkema'.

David R. Alkema
Laboratory Manager

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **110542**

Date Reported : **02/11/11**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Effluent"**

Sampled By : *AV of City of Cadillac, Water Resources*

Date Received : **02/08/11**

Sample Date : **02/07/11**

Sample Type : **aqueous**

Sample Time : **1019**

KAR Sample No. : **110542-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	02/08/11	PML	
Mercury by EPA 1631	0.6	ng/L	EPA 1631	02/09/11	PML	

Sample ID : **"Effluent, Field Duplicate"**

Sampled By : *AV of City of Cadillac, Water Resources*

Date Received : **02/08/11**

Sample Date : **02/07/11**

Sample Type : **aqueous**

Sample Time : **1021**

KAR Sample No. : **110542-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	02/08/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	02/09/11	PML	

Sample ID : **"Field Blank"**

Sampled By :

Date Received : **02/08/11**

Sample Date : **02/07/11**

Sample Type : **aqueous**

Sample Time :

KAR Sample No. : **110542-03**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	02/08/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	02/09/11	PML	

Sample ID : **"Trip Blank"**

Sampled By :

Date Received : **02/08/11**

Sample Date :

Sample Type : **Trip Blk**

Sample Time :

KAR Sample No. : **110542-04**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	02/08/11	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	02/09/11	PML	



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Attn : Ms. Amy Vail

KAR Project No. : 110539

Date Reported : 02/17/11

Date Activated : 02/08/11

Date Due : 02/22/11

Date Validated : 02/15/11

Project

Description : Analysis of one aqueous sample for Mercury Minimization.

Dear Client,

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If you wish to contact us about this work please mention KAR Project No. 110539. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,


David R. Alkema
Laboratory Manager

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **110539**

Date Reported : **02/17/11**

Attest:


David R. Alkema, Lab Manager

Project

Description: *Analysis of one aqueous sample for Mercury Minimization.*

Sample ID : **"Influent"**

Sampled By : *AV of City of Cadillac, Water Resources*

Date Received : **02/08/11**

Sample Date : **02/07/11**

Sample Type : **aqueous**

Sample Time : **0906**

KAR Sample No. : **110539-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
<i>Prep, Hg</i>	<i>Completed</i>		<i>EPA 245.2</i>	<i>02/11/11</i>	<i>DBL</i>	
<i>Mercury, total, low level</i>	<i><0.0002</i>	<i>mg/L</i>	<i>EPA 245.2</i>	<i>02/11/11</i>	<i>DBL</i>	